YU-HSUAN TSAI

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EDUCATION

National Taiwan University, Taipei, Taiwan — Sep 2026 (Expected)
M.S. in Computer Science (Incoming)

National Yang Ming Chiao Tung University, Hsinchu, Taiwan — Sep 2022 – Jun 2026
Bachelor of Computer Science

- **Major GPA:** 4.03 / 4.3 **Focus Areas:** Machine Learning, Software Engineering
- Relevant Coursework: Artificial Intelligence (A+), Machine Learning (A+), Software Engineering (A+)

TECHNICAL SKILLS

Programming	C, C++, Python, C#, JavaScript
ML / AI	PyTorch, TensorFlow, YOLO
Data & Analysis	NumPy, Pandas, data preprocessing, feature extraction, model evaluation
Software / Tools	Git, Unity (C#), macOS APIs, WebSocket
Strengths	Problem solving, debugging, system integration, team collaboration

EXPERIENCE

Work

- **Software Intern, Mechatronics Team — Logitech** — Mar 2026 – Present
  - Develop **Python-based analysis tools** for keyboard backlight evaluation, streamlining data processing and automation.
  - Implement **YOLO-based detection** models to support and enhance automated image defect analysis.
  - Execute **firmware and software integration** for development boards, including SDK and system-level debugging.
- **Research Assistant, Cognitive Engineering Lab** — 2024 – 2025
  - Built **EEG preprocessing and feature extraction pipelines** in Python, ensuring data consistency across sessions.
  - Optimized data flow between hardware sensors and processing units for real-time analysis.

Research / Applied Projects

- **XR Research — Unity Developer** — 2025
  - Developed C# experimental environments for **Meta Quest** focused on interaction precision and spatial design.
  - *Two projects Awarded Bronze and Special Presentation Awards at NYCU XR Showcase.*
  - **Portal**: Integrated virtual–physical interactions for a two-player asymmetric puzzle.
  - **Bio Hazard**: Implemented gesture-based interactions using **spatial scaling** and adaptive logic.
- **VR–BCI Rehabilitation System (EEG + Python + LLM)** — 2024 – 2025
  - Engineered an end-to-end pipeline combining **EEG signal processing** and **LLM-based feedback**.
  - Automated session analytics to provide real-time performance insights for motor-imagery tasks.
- **mac-network-monitor** [GitHub] — 2024
  - Developed a macOS tool for **real-time traffic visualization** using socket APIs and asynchronous I/O.

INTERESTS

- Building real-world software systems that integrate machine learning into user-facing products.
- Scalable system design, data-driven iteration, and improving reliability of production workflows.